

Exercises

Level - 1

(Problems Based on Fundamentals)

ABC OF AN ELIPSE

1. Find the centre, vertices, co-vertices, lengths of major and minor axes, eccentricity, lengths of latus rectum, equation of directrices and the end-points of a latus recta.
(i) $9x^2 + 16y^2 = 144$
(ii) $2x^2 + 3y^2 - 4x - 12 + 8 = 0$
2. Find the sum of the focal distance of any point on the ellipse $16x^2 + 25y^2 = 400$.
3. If the equation $\frac{x^2}{10-a} + \frac{y^2}{a-4} = 1$ represents of an ellipse such that the length of the interval, where a lies, is m, find m.
4. If (5, 12) and (24, 7) are the foci of an ellipse passing through the origin, find the eccentricity of the ellipse.
5. Find the equation of the ellipse whose axes are co-ordinate axes and foci are $(\pm 2, 0)$ and the eccentricity is $1/2$.
6. If the distance between the foci of an ellipse is equal to the length of its latus rectum, the eccentricity is $\frac{\sqrt{5}-1}{2}$.
7. Find the eccentricity of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ whose latus rectum is half of its major axis.
8. Find the eccentric angle of a point on the ellipse $\frac{x^2}{6} + \frac{y^2}{2} = 1$ whose distance from the centre of the ellipse is $\sqrt{2}$.
9. Find the area of the greatest rectangle that can be inscribed in the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.
10. Find the locus of a point whose co-ordinates are given by $x = 3 + 4 \cos \theta$ and $y = 2 + 3 \sin \theta$.
11. If PSQ is a focal chord of an ellipse $16x^2 + 25y^2 = 400$ such that $SP = 8$, find the length of SQ.

12. Find the area bounded by the curve $\frac{x^2}{16} + \frac{y^2}{9} \leq 1$ and the line $\frac{x}{4} + \frac{y}{3} \geq 1$.

POSITION OF A POINT W.R.T. AN ELLIPSE

13. Find the location of the point (2, 3) with respect to the ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$.
14. If $(\lambda, -\lambda)$ be an interior point of an ellipse $4x^2 + 5y^2 = 1$ such that the length of the interval, where λ lies, is m , where $m \in \mathbb{Q}^+$, find the value of $(3m - 2)^{2013} + 2013$.

TANGENT AND TANGENCY

15. Find the number of tangents drawn from a point (2, 3) to an ellipse $4x^2 + 3y^2 = 12$.
16. Find the equations of the tangents drawn from the point (2, 3) to the ellipse $9x^2 + 16y^2 = 144$.
17. If the line $3x + 4y = 5$ touches the ellipse $9x^2 + 16y^2 = 144$, find the points of contact.
18. For what value of λ does the line $y = x + \lambda$ touches the ellipse $9x^2 + 16y^2 = 144$?
19. Find the equations of the tangents to the ellipse $\frac{x^2}{3} + \frac{y^2}{4} = 1$ having slope 2.
20. Prove that the locus of the feet of perpendicular drawn from the centre upon any tangent to the given ellipse is $r^2 = a^2 \cos^2 \theta + b^2 \sin^2 \theta$.
21. A circle of radius r is concentric with the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Prove that the common tangent is inclined to the major axis at an angle of $\tan^{-1} \left(\sqrt{\frac{r^2 - b^2}{a^2 - r^2}} \right)$.
22. Prove that the tangents at the extremities of latus rectum of an ellipse intersect on the corresponding directrix.
23. Prove that the locus of the mid-points of the portion of the tangents to the given ellipse intercepted between the axes is $4r^2 = a^2 \sin^2 \theta + b^2 \cos^2 \theta$.
24. A tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ meets the ellipse $\frac{x^2}{a} + \frac{y^2}{b} = a + b$ in the points P and Q , prove that the tangents at P and Q are at right angles.
25. Prove that the product of the perpendiculars drawn from the foci upon any tangent to an ellipse is constant, i.e. b^2 .
26. If an ellipse slides between two perpendicular straight lines, prove that the locus of its centre is a circle.
27. Prove that the locus of the feet of the perpendiculars from foci upon any tangent to an ellipse is an auxiliary circle.

28. If p be the length of perpendicular from the focus S of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ on the tangent at P , prove that

$$\frac{b^2}{p^2} = \frac{2a}{SP} - 1.$$

29. If p be the perpendicular from the centre of an ellipse upon the tangent at any point P on it and r be the distance of P from the centre, prove that $\frac{a^2 b^2}{p^2} = a^2 + b^2 - r^2$.

30. Prove that the locus of the mid-points of the portion of the tangents to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ intercepted between the axes is $\frac{a^2}{x^2} + \frac{b^2}{y^2} = 4$.

31. Prove that the portion of the tangent to the ellipse intercepted between the curve and the directrix subtends a right angle at the corresponding focus.

32. Prove that in an ellipse, the perpendicular from a focus upon any tangent and the line joining the centre of the ellipse to the point of contact meet on the corresponding directrix.

33. Prove that the co-ordinates of those points on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ tangents at which make equal angles with the axes is $\left(\pm \frac{a^2}{\sqrt{a^2 + b^2}}, \pm \frac{b^2}{\sqrt{a^2 + b^2}} \right)$.

34. Find the locus of the point of intersection of two perpendicular tangents to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

35. Tangents are drawn from any point P on the parabola $(y - 2)^2 = 4(x - 1)$ to the ellipse $\frac{x^2}{4} + \frac{y^2}{1} = 1$, which are mutually perpendicular to each other. Find the locus of the point P .

36. Find the equations of the pair of tangents to the ellipse $2x^2 + 3y^2 = 1$ from the point (1, 1).

37. If the tangents are drawn from a point (1, 2) to the ellipse $3x^2 + 2y^2 = 5$, find the angle between the tangents.

NORMAL AND NORMALCY

38. Find the equation of the normal to the ellipse $4x^2 + 9y^2 = 20$ at $\left(1, \frac{4}{3} \right)$.

39. Find the equation of the normal to the ellipse $5x^2 + 3y^2 = 137$ at the point whose ordinate is 2.

40. Find the equation of the normal to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ at the negative end of the latus rectum.

41. If the normal at the point $P(\theta)$ to the ellipse $5x^2 + 14y^2 = 70$ intersects it again at the point $Q(2\theta)$, prove that $3 \cos \theta + 2 = 0$.
42. The normal at an end of a latus rectum of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ passes through one extremity of the minor axis, then prove that $e^4 + e^2 - 1 = 0$.
43. Prove that the tangent and the normal at any point of an ellipse bisect the the external and internal angles between the focal distances of the point.
44. The normal at any point P on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ meets the major and minor axes at G and G' , respectively and CF is perpendicular upon the normal from the centre C of the ellipse, show that $PF \cdot PG = b^2$ and $PF \times PG' = a^2$.
45. The normal at a point $P(\theta)$ on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ meets the axes of x and y at M and N , respectively, show that $x^2 + y^2 = (a + b)^2$.
46. An ordinate PN of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ meets the auxiliary circle in Q . Show that the locus of the point of intersection of the normals at P and Q is the circle $x^2 + y^2 = (a + b)^2$.
47. Prove that the tangent of the angle between CP and the normal at $P(\theta)$ is $\left(\frac{a^2 - b^2}{2ab}\right) \times \sin 2\theta$ and its greatest value is $\left(\frac{a^2 - b^2}{2ab}\right)$.
48. Prove that in an ellipse, the distance between the centre and any normal does not exceed the difference between the semi-axes of the curve.
49. The tangent and the normal at any point P of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ cut its major axis in points Q and R , respectively. If $QR = a$, show that the eccentric angle of the point P is satisfying the equation $e^2 \cos^2 \phi + \cos \phi - 1 = 0$.
50. If the normals at $P(x_1, y_1)$, $Q(x_2, y_2)$ and $R(x_3, y_3)$ on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ are concurrent, show that

$$\begin{vmatrix} x_1 & y_1 & x_1 y_1 \\ x_2 & y_2 & x_2 y_2 \\ x_3 & y_3 & x_3 y_3 \end{vmatrix} = 0$$

and if points $P(\alpha)$, $Q(\beta)$ and $R(\gamma)$, prove that

$$\begin{vmatrix} \sec \alpha & \operatorname{cosec} \alpha & 1 \\ \sec \beta & \operatorname{cosec} \beta & 1 \\ \sec \gamma & \operatorname{cosec} \gamma & 1 \end{vmatrix} = 0$$

CHORD OF CONTACT/CHORD BISECTED AT A POINT

51. Prove that the locus of the point, the chord of contact of tangents from which to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ subtends a right angle at the centre of the ellipse is $\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{a^2} + \frac{1}{b^2}$.
52. Prove that the locus of the point, from which the chord of contact of tangents are to be drawn to the ellipse touches the circle $x^2 + y^2 = c^2$ is $\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{c^2}$.
53. The perpendicular tangents are drawn to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Prove that the locus of the mid-point of the chord of contact is $\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^2 = \left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)$.
54. Tangents are drawn from any point on the circle $x^2 + y^2 = c^2$ to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Prove that the locus of the mid-points of the chord of contact is $\left(\frac{x^2}{c^2} + \frac{y^2}{b^2}\right)^2 = \left(\frac{x^2}{c^2} + \frac{y^2}{b^2}\right)$.
55. Tangents PA and PB are drawn from a point P to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The area of the triangle formed by the chord of contact AB and axes of co-ordinates are constant. Prove that the locus of P is a hyperbola.
56. Prove that the locus of the mid-points of the normal chords of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\left(\frac{a^6}{x^2} + \frac{b^6}{y^2}\right) \left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^2 = (a^2 - b^2)^2$.
57. Prove that the locus of the mid-points of the chords of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ which touch the auxiliary circle $x^2 + y^2 = a^2$ is $a^2 \left(\frac{x^2}{a^4} + \frac{y^2}{b^4}\right) = \left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^2$.
58. Prove that the locus of the mid-points of the chords of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ which subtends a right angle at the centre of the ellipse is $\left(\frac{x^2}{a^4} + \frac{y^2}{b^4}\right) = \left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right) \left(\frac{1}{a^2} + \frac{1}{b^2}\right)$.
59. The eccentric angles of two points P and Q on the ellipse differ by $\pi/2$. Prove that the locus of the mid-point of PQ is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$.

60. Prove that the locus of the mid-point of the chord of contact of the perpendicular tangents to the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ is } \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right)^2 = \left(\frac{x^2 + y^2}{a^2 + b^2} \right).$$

61. Prove that the locus of the mid-points of the focal chords of the ellipse $\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right) = 1$ is
- $$b^2x^2 + a^2y^2 = ab^2xe$$

62. Prove that the locus of the mid-points of the chords of the ellipse $\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right) = 1$ which are tangents to the ellipse $\frac{x^2}{\alpha^2} + \frac{y^2}{\beta^2} = 1$ is

$$\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right)^2 = \left(\frac{x^2 \alpha^2}{a^4} + \frac{y^2 \beta^2}{b^4} \right)$$

POLE AND POLAR

63. Prove that the polar of the focus of an ellipse is the directrix.
64. Find the pole of a given line $lx + my + n = 0$ with respect to an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.
65. Prove that the pole of a given line is the same as the point of intersection of tangents at its extremities.
66. Find the pole of the straight line $x + 4y = 4$ with respect to the ellipse $x^2 + 4y^2 = 4$.
67. Find the locus of the poles of the tangents to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with respect to the concentric ellipse $c^2x^2 + d^2y^2 = 1$.
68. The perpendicular from the centre of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ on the polar of a point with respect to the ellipse is constant and equal to c . Prove that the locus of the point is the ellipse $\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{c^2}$.
69. Show that the equation of the locus of the poles of normal chords of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is
- $$(a^2 - b^2)^2 x^2 y^2 = a^6 y^2 + b^6 x^2$$
70. If the polar with respect to $y^2 = 4x$ touches the ellipse $\frac{x^2}{\alpha^2} + \frac{y^2}{\beta^2} = 1$, find the locus of its pole.

CONJUGATE DIAMETERS

71. Prove that the equation of the diameter of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $y = -\frac{b^2x}{a^2m}$.

72. Find the co-ordinates of the four extremities of two conjugate diameters of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

73. Prove that the sum of the squares of any two conjugate semi-diameters of an ellipse is constant and equal to the sum of the squares of the semi-axes of the ellipse, i.e. $CP^2 + CD^2 = a^2 + b^2$.

74. Prove that the product of the focal distances of a point on an ellipse is equal to the square of the semi-diameter, which is conjugate to the diameter through the point. i.e. $SP \cdot S'P = CD^2$.

75. Prove that the locus of the poles of the line joining the eccentricities of two conjugate diameters is the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$.

76. If P and D be the ends of the conjugate diameters of an ellipse, find the locus of the mid-point of PD .

77. For the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, find the equation of the diameter conjugate to $ax - by = 0$.

78. If the point of intersection of the ellipses $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $\frac{x^2}{c^2} + \frac{y^2}{d^2} = 1$ be at the extremities of the conjugate diameters of the former, prove that $\frac{a^2}{c^2} + \frac{b^2}{d^2} = 1$.

79. If CP and CD are the conjugate diameters of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, prove that the locus of the orthocentre of the ΔPCD is $2(b^2y^2 + a^2x^2)^2 = (a^2 - b^2)^2 (b^2y^2 - a^2x^2)^2$.

80. Prove that the tangents at the extremities of a pair of conjugate diameters form a parallelogram whose area is constant and equal to the product of the axes.

81. Show that the tangents at the ends of conjugate diameters of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ intersect on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$.

82. Find the eccentricity of the ellipse if $y = x$ and $2x + 3y = 0$ are the equations of a pair of its conjugate diameters.

REFLECTION PROPERTY OF AN ELLIPSE

83. A ray is emanating from the point $(-3, 0)$ is incident on the ellipse $16x^2 + 25y^2 = 400$ at the point P with ordinate 4. Find the equation of the reflected ray after first reflection.
84. A ray is coming along the line $x - y + 2 = 0$ on the ellipse $3x^2 + 4y^2 = 12$. After striking the elliptic mirror, it is then reflected. Find the equation of the line containing the reflected ray.

HINTS AND SOLUTIONS

LEVEL I

1. (i) The given equation of the ellipse is $9x^2 + 16y^2 = 144$

$$\Rightarrow \frac{x^2}{16} + \frac{y^2}{9} = 1$$

- (a) Centre: $C(0, 0)$
 (b) Vertices: $A(a, 0)$ $A'(-a, 0) = A(4, 0)$ and $A'(4, 0)$
 (c) Co-vertices: $B = (0, b)$ and $B' = (0, -b)$
 $\Rightarrow B = (0, 3)$ and $B' = (0, -3)$
 (d) Length of the major axis: $2a = 8$
 (e) Length of the minor axis: $2b = 6$

(f) Eccentricity $= e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{9}{16}} = \frac{\sqrt{7}}{4}$

(g) Lengths of the latus rectum $= \frac{2b^2}{a} = \frac{18}{4} = \frac{9}{2}$

- (h) Equations of the directrices:

$$x = \pm \frac{a}{e}$$

$$\Rightarrow x = \pm \frac{4}{\frac{\sqrt{7}}{4}} = \pm \frac{16}{\sqrt{7}}$$

- (i) End-points of the latus recta:

$$L\left(ae, \frac{b^2}{a}\right), L'\left(ae, -\frac{b^2}{a}\right)$$

$$= L\left(\sqrt{7}, \frac{9}{4}\right), L'\left(\sqrt{7}, -\frac{9}{4}\right)$$

- (ii) The given equation of the ellipse is

$$2x^2 + 3y^2 - 4x - 12y + 8 = 0$$

$$\Rightarrow 2(x^2 - 2x + 1) + 3(y^2 - 4y + 4) = 6$$

$$\Rightarrow 2(x-1)^2 + 3(y-2)^2 = 6$$

$$\Rightarrow \frac{(x-1)^2}{3} + \frac{(y-2)^2}{2} = 1$$

$$\Rightarrow \frac{X^2}{3} + \frac{Y^2}{2} = 1, \text{ where } X = x-1, Y = y-2$$

- (a) Centre: $(0, 0)$
 $\Rightarrow X = 0, Y = 0$
 $\Rightarrow x-1 = 0$ and $y-2 = 0$
 $\Rightarrow x = 1$ and $y = 2$
 Hence, the centre is $(1, 2)$.

- (b) Vertices: $(\pm a, 0)$
 $\Rightarrow X = \pm a, Y = 0$
 $\Rightarrow x-1 = \pm 3, y-2 = 0$
 $\Rightarrow x = 1 \pm 3, y = 2$
 $\Rightarrow x = 4, -2, y = 2$
 Hence, the vertices are $(4, 2)$ and $(-2, 2)$

- (c) Co-vertices: $(0, \pm b)$

$$\Rightarrow X = 0, Y = \pm 3$$

$$\Rightarrow x-1 = 0, y-2 = \pm 3$$

$$\Rightarrow x = 1, y = 2 \pm 3$$

$$\Rightarrow x = 1, y = 5, -1$$

Hence, the co-vertices are $(1, 5), (1, -1)$.

- (d) The length of the major axis $= 2a = 6$

- (e) The length of the minor axis $= 2b = 4$

- (f) The length of the latus rectum $=$

$$\frac{2b^2}{a} = \frac{18}{4} = \frac{9}{2}$$

- (g) Eccentricity,

$$e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{2}{3}} = \frac{1}{\sqrt{3}}$$

- (h) Equations of the directrices:

$$X = \pm \frac{a}{e} = \pm \frac{3}{1/\sqrt{3}} = \pm 3\sqrt{3}$$

$$\Rightarrow x = 1 \pm 3\sqrt{3}$$

- (i) End-points of the latus recta:

$$L\left(ae, \frac{b^2}{a}\right); L'\left(ae, -\frac{b^2}{a}\right)$$

$$= L\left(\sqrt{3}, \frac{9}{2}\right); L'\left(\sqrt{3}, -\frac{9}{2}\right)$$

2. The given equation of the ellipse is

$$16x^2 + 25y^2 = 400$$

$$\Rightarrow \frac{x^2}{25} + \frac{y^2}{16} = 1$$

Thus, the sum of the focal distances

$$= 2a = 10 \text{ units}$$

3. The given equation of an ellipse is $\frac{x^2}{10-a} + \frac{y^2}{a-4} = 1$

$$\Rightarrow (a-4)x^2 + (10-a)y^2 - (a-4)(10-a) = 0$$

Since, the given equation represents an ellipse,

$$h^2 - ab < 0$$

$$\Rightarrow 0 - (a-4)(10-a) < 0$$

$$\Rightarrow (a-4)(a-10) < 0$$

$$\Rightarrow 4 < a < 10$$

Hence, the length of the interval $= 10 - 4 = 6$.

Thus, the value of m is 6.

4. Let the foci of the ellipse are S and S' respectively.

Then $S = (5, 12)$ and $S' = (24, 7)$.

Thus, the centre of the ellipse is $C\left(\frac{29}{2}, \frac{19}{2}\right)$.

$$\text{Now } OC = \frac{1}{2} \times \sqrt{29^2 + 19^2} = \frac{1}{2} \times \sqrt{1202}$$

We have,

$$SS' = \sqrt{386}$$

$$\Rightarrow e = \frac{\sqrt{386}}{2a} = \frac{\sqrt{386}}{\sqrt{1202}} = \sqrt{\frac{386}{1202}} = \frac{1}{\sqrt{3}}$$

5. Let $S(2, 0)$ and $S'(-2, 0)$ are two foci of the ellipse and $C(0, 0)$ be the centre of the ellipse.

We have,

$$SS' = 4$$

$$\Rightarrow 2ae = 4$$

$$\Rightarrow 2a \times \frac{1}{2} = 4 \Rightarrow a = 4$$

$$\text{Also, } b^2 = a^2(1 - e^2) = 16\left(1 - \frac{1}{4}\right) = 12$$

$$\Rightarrow b = \sqrt{12} = 2\sqrt{3}$$

Hence, the equation of the ellipse is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\Rightarrow \frac{x^2}{16} + \frac{y^2}{12} = 1$$

6. Given $2ae = \frac{2b^2}{a}$

$$\Rightarrow b^2 = a^2e$$

$$\Rightarrow a^2(1 - e^2) = a^2e$$

$$\Rightarrow (1 - e^2) = e$$

$$\Rightarrow e^2 + e - 1 = 0$$

$$\Rightarrow e = \frac{\sqrt{5} - 1}{2}$$

Hence, the result

7. The equation of the given ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

It is given that the length of the latus rectum = $\frac{1}{2} \times$ the length of the major axis.

$$\Rightarrow \frac{2b^2}{a} = a$$

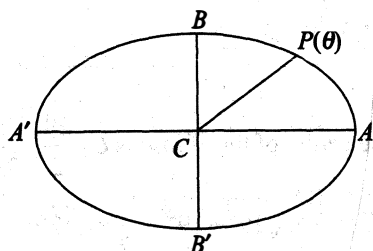
$$\Rightarrow b = a$$

As we know that,

$$\text{eccentricity} = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - 1} = 0$$

8. Let any point on the ellipse $\frac{x^2}{6} + \frac{y^2}{2} = 1$ is

$P(\sqrt{6} \cos \theta, \sqrt{2} \sin \theta)$ and C be the centre of the ellipse.



Therefore,

$$CP = \sqrt{5}$$

$$\Rightarrow 6 \cos^2 \theta + 2 \sin^2 \theta = 5$$

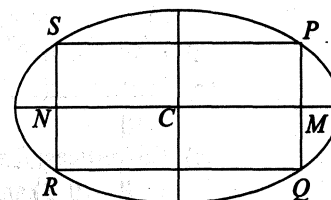
$$\Rightarrow 4 \cos^2 \theta = 3$$

$$\Rightarrow \cos \theta = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}$$

Hence, the eccentric angle is $\frac{\pi}{6}$.

9. Let Q be any point on the given ellipse, whose co-ordinates are $(4 \cos^2 \theta, 3 \sin^2 \theta)$.



Thus, $PQ = 8 \cos \theta$

and $QR = 6 \sin \theta$

Thus, the area of the rectangle PQRS

$$= PQ \times QR$$

$$= 48 \sin \theta \cos \theta$$

$$= 24 \sin 2\theta$$

Hence, the area of the greatest rectangle

$$= 24 \text{ sq.u. at } \theta = \frac{\pi}{4}$$

10. The co-ordinates of the given point are $x = 3 + 4 \cos \theta, y = 2 + 3 \sin \theta$

$$\Rightarrow \frac{(x-3)}{4} = \cos \theta, \frac{(y-2)}{3} = \sin \theta$$

$$\Rightarrow \frac{(x-3)^2}{4^2} + \frac{(y-2)^2}{3^2} = \cos^2 \theta + \sin^2 \theta = 1$$

$$\Rightarrow \frac{(x-3)^2}{16} + \frac{(y-2)^2}{9} = 1$$

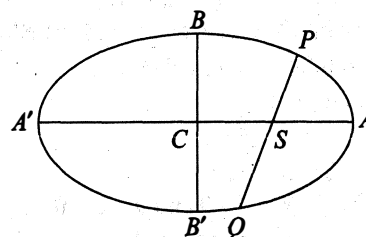
which is the required locus.

11. The equation of the ellipse is

$$16x^2 + 25y^2 = 400$$

$$\Rightarrow \frac{x^2}{25} + \frac{y^2}{16} = 1$$

As we know that, if SP and SQ are the focal segments of a focal chord PSQ , then $\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a}$



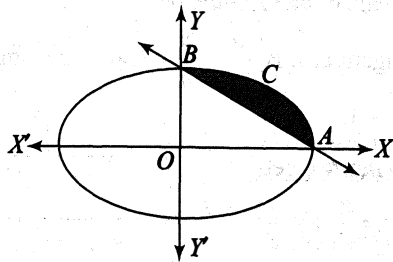
$$\Rightarrow \frac{1}{8} + \frac{1}{SQ} = \frac{1}{5}$$

$$\Rightarrow \frac{1}{SQ} = \frac{1}{5} - \frac{1}{8} = \frac{8-5}{40} = \frac{3}{40}$$

$$\Rightarrow SQ = \frac{40}{3}$$

Hence, the length of SQ is $40/3$.

12. As we know that the area of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab .



Thus, the area of $OACBO$

$$= \left(\frac{3}{4} \times \pi \cdot 4 \cdot 3 - \frac{1}{2} \cdot 4 \cdot 3 \right) \\ = (9\pi - 6) \text{ s.u.}$$

13. Let $N = \frac{x_1^2}{4} + \frac{y_1^2}{3} - 1$

Then, $N = \frac{4}{4} + \frac{9}{3} - 1 = 1 + 3 - 1 = 3$

Since, the value of N is positive at $(2, 3)$, so the point $(2, 3)$ lies outside of the ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$.

14. Since the point $(\lambda, -\lambda)$ be an interior point of an ellipse $4x^2 + 5y^2 = 1$, then

$$4\lambda^2 + 5\lambda^2 - 1 < 0 \\ \Rightarrow 9\lambda^2 - 1 < 0 \\ \Rightarrow (3\lambda + 1)(3\lambda - 1) < 0 \\ \Rightarrow -\frac{1}{3} < \lambda < \frac{1}{3}$$

Therefore, $m = \frac{2}{3}$

$$\Rightarrow (3m - 2)^{2013} + 2013 = 0 + 2013 = 2013$$

15. We have,

$$4 \cdot 4 + 3 \cdot 3 - 12 = 16 + 9 - 12 = 13 > 0$$

Thus, the point $(2, 3)$ lies outside of the ellipse.

Thus, the number of tangents = 2

16. The given ellipse is

$$9x^2 + 16y^2 = 144$$

$$\Rightarrow \frac{x^2}{16} + \frac{y^2}{9} = 1$$

The equation of any tangent to the given ellipse is

$$y = mx + \sqrt{a^2 m^2 + b^2}$$

$$\Rightarrow y = mx + \sqrt{16m^2 + 9}$$

which is passing through $(2, 3)$, so

$$3 = 2m + \sqrt{16m^2 + 9}$$

$$\Rightarrow (3 - 2m)^2 = 16m^2 + 9$$

$$\Rightarrow 9 - 6m + 4m^2 = 16m^2 + 9$$

$$\Rightarrow 12m^2 + 6m = 0$$

$$\Rightarrow 2m^2 + m = 0$$

$$\Rightarrow m(2m + 1) = 0$$

$$\Rightarrow m = 0, -\frac{1}{2}$$

Hence, the equation of tangents are $y = 3$ and

$$y = -\frac{1}{2}x + \sqrt{13}.$$

17. The given line is

$$3x + 4y = 5$$

$$\Rightarrow 4y = -3x + 5$$

and the ellipse is

$$9x^2 + 16y^2 = 144$$

$$\Rightarrow y = -\frac{3}{4}x + \frac{5}{4} \text{ and } \frac{x^2}{16} + \frac{y^2}{9} = 1$$

As we know that the line $y = mx + c$ will be a tangent

to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, the points of contacts are

$$\left(\pm \frac{a^2 m}{c}, \pm \frac{b^2}{c} \right)$$

$$= \left(\pm \frac{16 \left(-\frac{3}{4} \right)}{\frac{5}{4}}, \pm \frac{9}{\frac{5}{4}} \right) \\ = \left(\pm \frac{(-48)}{5}, \pm \frac{36}{5} \right)$$

18. The equation of the given ellipse is

$$9x^2 + 16y^2 = 144$$

$$\Rightarrow \frac{x^2}{16} + \frac{y^2}{9} = 1$$

The line $y = x + \lambda$ will be a tangent to the ellipse, if

$$c^2 = a^2 m^2 + b^2$$

$$\Rightarrow \lambda^2 = 16 \cdot 1 + 9 = 25$$

$$\Rightarrow \lambda = \pm 5$$

Hence, the values of λ are ± 5 .

19. The given ellipse is $\frac{x^2}{3} + \frac{y^2}{4} = 1$

The equation of any ellipse is

$$y = mx \pm \sqrt{a^2 m^2 + b^2}$$

$$\Rightarrow y = 2x \pm \sqrt{3 \cdot 4 + 2} = 2x \pm \sqrt{14}$$

20. The equation of any tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at $P(\theta)$ is

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1 \quad \dots(i)$$

The equation of perpendiculars from centre $(0, 0)$ to tangent is

$$\frac{x}{b} \sin \theta - \frac{y}{a} \cos \theta = 0 \quad \dots(ii)$$

From Eq. (ii), we get

$$\frac{\sin \theta}{by} = \frac{\cos \theta}{ax} = \frac{1}{\sqrt{a^2 x^2 + b^2 y^2}} \quad \dots(iii)$$

The locus of the feet of perpendiculars is the point of intersection of (i) and (ii).

It is obtained by eliminating θ between Eqs (i) and (ii).

Squaring Eqs (i) and (ii) and adding, we get

$$\frac{x^2}{a^2} \cos^2 \theta + \frac{y^2}{b^2} \sin^2 \theta + \frac{x^2}{b^2} \sin^2 \theta + \frac{y^2}{a^2} \cos^2 \theta = 1$$

$$\Rightarrow \left(\frac{x^2}{a^2} + \frac{y^2}{a^2} \right) \cos^2 \theta + \left(\frac{y^2}{b^2} + \frac{x^2}{b^2} \right) \sin^2 \theta = 1$$

$$\Rightarrow (x^2 + y^2) \left(\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2} \right) = 1$$

$$\Rightarrow (x^2 + y^2)(b^2 \cos^2 \theta + a^2 \sin^2 \theta) = a^2 b^2$$

$$\Rightarrow (x^2 + y^2) \left(\frac{b^2 a^2 x^2}{a^2 x^2 + b^2 y^2} + \frac{b^2 a^2 y^2}{a^2 x^2 + b^2 y^2} \right) = a^2 b^2$$

$$\Rightarrow (x^2 + y^2)^2 = (a^2 x^2 + b^2 y^2)$$

Now, put $x = r \cos \theta$ and $y = r \sin \theta$, we get

$$r^2 = a^2 \cos^2 \theta + b^2 \sin^2 \theta.$$

Hence, the result.

21. The equation of circle is $x^2 + y^2 = r^2$.

So the equation of any tangent to the ellipse is

$$y = mx + \sqrt{a^2 m^2 + b^2}$$

If it is a tangent to a circle also, then the length of the perpendicular from the centre $(0, 0)$ of a circle is equal to the radius of a circle.

$$\text{Thus, } \left| \frac{\sqrt{a^2 m^2 + b^2}}{\sqrt{1 + m^2}} \right| = r$$

$$\Rightarrow a^2 m^2 + b^2 = r^2 (1 + m^2) = r^2 + r^2 m^2$$

$$\Rightarrow (a^2 - r^2) m^2 = (r^2 - b^2)$$

$$\Rightarrow m^2 = \left(\frac{r^2 - b^2}{a^2 - r^2} \right)$$

$$\Rightarrow m = \sqrt{\left(\frac{r^2 - b^2}{a^2 - r^2} \right)}$$

$$\Rightarrow \tan \theta = \sqrt{\left(\frac{r^2 - b^2}{a^2 - r^2} \right)}$$

$$\Rightarrow \theta = \tan^{-1} \left(\sqrt{\left(\frac{r^2 - b^2}{a^2 - r^2} \right)} \right)$$

Hence, the result.

22. Let the end-points of a latus rectum are

$$L(ae, 0) \text{ and } L(-ae, 0).$$

The tangent at L is $\frac{xe}{a} + \frac{y}{a} = 1$ and the tangent at L' is $\frac{xe}{a} - \frac{y}{a} = 1$.

On solving, we get,

$$x = \frac{a}{e} \text{ and } y = 0.$$

$\Rightarrow x = \frac{a}{e}$ is the equation of directrix to the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

23. The equation of any tangent to the ellipse at $P(\theta)$ is

$$AB: \frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1.$$

Thus $A = (a \sec \theta, 0)$ and $A = (0, b \csc \theta)$

Let (h, k) be the mid-point of the tangent AB .

$$\text{Therefore, } h = \frac{a \sec \theta}{2} \text{ and } k = \frac{b \csc \theta}{2}$$

$$\Rightarrow \cos \theta = \frac{a}{2h} \text{ and } \sin \theta = \frac{b}{2k}$$

Now squaring and adding, we get

$$\frac{a^2}{4h^2} + \frac{b^2}{4k^2} = 1$$

$$\Rightarrow \frac{a^2}{h^2} + \frac{b^2}{k^2} = 4$$

Hence the locus of (h, k) is

$$\frac{a^2}{x^2} + \frac{b^2}{y^2} = 4$$

Now, put $x = r \cos \theta$ and $y = r \sin \theta$, we get,

$$4r^2 = a^2 \sin^2 \theta + b^2 \cos^2 \theta,$$

Hence, the result.

24. Given ellipses are $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \dots(i)$

$$\text{and } \frac{x^2}{a} + \frac{y^2}{b} = a + b$$

$$\Rightarrow \frac{x^2}{a(a+b)} + \frac{y^2}{b(a+b)} = 1 \quad \dots(ii)$$

Let $R(h, k)$ be the points of intersection of the tangents to the ellipse (ii) at P and Q . Then PQ will be the chord of contact.

Thus its equation will be

$$\frac{hx}{a(a+b)} + \frac{ky}{b(a+b)} = 1$$

$$\Rightarrow y = -\frac{bh}{ak}x + \frac{b(a+b)}{k} \quad \dots(iii)$$

Since the line (iii) is a tangent to the ellipse (i), so we have

$$\begin{aligned} \frac{b^2(a+b)^2}{k^2} &= a^2 \times \frac{b^2}{a^2} \times \frac{h^2}{k^2} + b^2 \\ \Rightarrow b^2(a+b)^2 &= b^2h^2 + b^2k^2 \\ \Rightarrow h^2 + k^2 &= (a+b)^2 \end{aligned}$$

Thus the locus of (h, k) is

$$x^2 + y^2 = (a+b)^2$$

which is the director circle of the ellipse

$$\frac{x^2}{a} + \frac{y^2}{b} = a + b$$

Hence, the tangents at P and Q are at right angles.

25. Let two foci of an ellipse are

$$S(ae, 0) \text{ and } S(-ae, 0).$$

The equation of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at $P(\theta)$ is $\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$.

Let p_1 and p_2 be the lengths of perpendiculars from the given foci to the given ellipse.

$$\text{Thus, } p_1 = \left| \frac{1 - e \cos \theta}{\sqrt{\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2}}} \right|$$

$$\text{and } p_2 = \left| \frac{1 + e \cos \theta}{\sqrt{\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2}}} \right|$$

$$\begin{aligned} \Rightarrow p_1 \times p_2 &= \left| \frac{1 - e \cos \theta}{\sqrt{\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2}}} \right| \left| \frac{1 + e \cos \theta}{\sqrt{\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2}}} \right| \\ &= \left(\frac{1 - e^2 \cos^2 \theta}{\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2}} \right) \\ &= \left(\frac{a^2 b^2 (1 - e^2 \cos^2 \theta)}{b^2 \cos^2 \theta + a^2 \sin^2 \theta} \right) \\ &= \left(\frac{a^2 b^2 (1 - e^2 \cos^2 \theta)}{a^2 (1 - e^2) \cos^2 \theta + a^2 \sin^2 \theta} \right) \\ &= b^2 \end{aligned}$$

26. Consider P is the point of intersection of two perpendicular tangents. Thus the locus of P is the director circle.

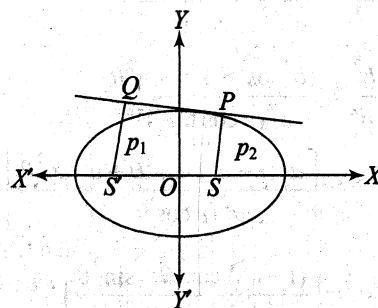
Therefore, the equation of a director circle is

$$x^2 + y^2 = a^2 + b^2$$

This means that the centre of the ellipse will always remain at a constant distance $\sqrt{a^2 + b^2}$ from P .

Hence, the locus of the centre is a circle.

27. The equation of any tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at $P(\theta)$ is



$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1 \quad \dots(i)$$

The equation of perpendiculars from foci $(\pm ae, 0)$ to the tangent is

$$\frac{x}{b} \sin \theta - \frac{y}{a} \cos \theta = \pm \frac{ae \sin \theta}{b} \quad \dots(ii)$$

Locus of the feet of perpendiculars is the point of intersection of (i) and (ii).

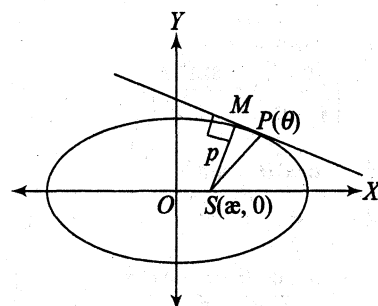
It is obtained by eliminating θ between Eqs (i) and (ii).

Squaring Eqs (i) and (ii) and adding, we get

$$\begin{aligned} \Rightarrow x^2 \left(\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2} \right) + y^2 \left(\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2} \right) &= 1 + \frac{a^2 e^2 \sin^2 \theta}{b^2} \\ \Rightarrow (x^2 + y^2) \left(\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2} \right) &= a^2 \left(\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2} \right) \\ \Rightarrow (x^2 + y^2) &= a^2 \end{aligned}$$

Hence, the result.

28. The tangent at $P(\theta)$ is $\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$.



Also $SP = a - ex = a(1 - e \cos \theta)$

Since p is the length of perpendicular from the focus $S(ae, 0)$,

$$\begin{aligned} \text{then, } p &= \frac{e \cos \theta - 1}{\sqrt{\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2}}} \\ \Rightarrow \frac{1}{p^2} &= \frac{(b^2 \cos^2 \theta + a^2 \sin^2 \theta)}{a^2 b^2 (e \cos \theta - 1)^2} \\ \Rightarrow \frac{b^2}{p^2} &= \frac{(b^2 \cos^2 \theta + a^2 \sin^2 \theta)}{a^2 (e \cos \theta - 1)^2} \\ &= \left[\frac{a^2 (1 - e^2) \cos^2 \theta + a^2 \sin^2 \theta}{a^2 (e \cos \theta - 1)^2} \right] \\ &= \left[\frac{(1 - e^2) \cos^2 \theta + \sin^2 \theta}{(e \cos \theta - 1)^2} \right] \\ &= \left[\frac{(1 - e^2 \cos^2 \theta)}{(e \cos \theta - 1)^2} \right] \\ &= \left[\frac{1 + e \cos \theta}{1 - e \cos \theta} \right] \end{aligned}$$

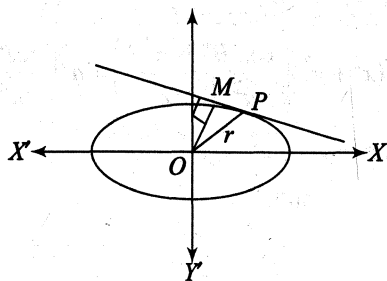
Also,

$$\frac{2a}{SP} - 1 = \frac{2a}{a(1 - e \cos \theta)} - 1 = \frac{1 + e \cos \theta}{1 - e \cos \theta}$$

$$\text{Hence, } \frac{b^2}{p^2} = \frac{2a}{SP} - 1$$

29. Let the point P be $(a \cos \theta, b \sin \theta)$.
The equation of the tangent at P be

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$$



$$\begin{aligned} \text{Now, } p &= \frac{0 + 0 - 1}{\sqrt{\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2}}} \\ \Rightarrow \frac{1}{p^2} &= \frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2} \\ &= \frac{a^2 \sin^2 \theta + b^2 \cos^2 \theta}{a^2 b^2} \end{aligned}$$

$$\begin{aligned} \Rightarrow \frac{a^2 b^2}{p^2} &= a^2 \sin^2 \theta + b^2 \cos^2 \theta \\ &= a^2 (1 - \cos^2 \theta) + b^2 (1 - \sin^2 \theta) \\ &= a^2 + b^2 - (a^2 \cos^2 \theta + b^2 \sin^2 \theta) \\ &= a^2 + b^2 - r^2 \end{aligned}$$

Hence, the result.

30. Any tangent to the ellipse

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$$

Let it meet the x -axis at A and y -axis at B . Then the co-ordinates of A and B are

$$A = \left(\frac{a}{\cos \theta}, 0 \right) \text{ and } B = \left(0, \frac{b}{\sin \theta} \right)$$

Let $M(h, k)$ be the mid-point of AB .

$$\text{Then } 2h = \frac{a}{\cos \theta} \text{ and } 2k = \frac{b}{\sin \theta}$$

$$\Rightarrow h = \frac{a}{2 \cos \theta} \text{ and } k = \frac{b}{2 \sin \theta}$$

$$\Rightarrow \cos \theta = \frac{a}{2h} \text{ and } \sin \theta = \frac{b}{2k}$$

Squaring and adding, we get

$$\begin{aligned} \frac{a^2}{4h^2} + \frac{b^2}{4k^2} &= 1 \\ \Rightarrow \frac{a^2}{h^2} + \frac{b^2}{k^2} &= 4 \end{aligned}$$

Hence, the locus of $M(h, k)$ is

$$\frac{a^2}{x^2} + \frac{b^2}{y^2} = 4$$

31. The tangent at $P(a \cos \theta, b \sin \theta)$ is

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$$

Let it meet the directrix $x = \frac{a}{e}$ at Q , where Q is

$$\left(\frac{a}{e}, \frac{b(e - \cos \theta)}{e \sin \theta} \right)$$

$$\text{Slope of } SP = m_1 = \frac{b(e - \cos \theta)}{e \sin \theta}$$

$$\begin{aligned} \text{Slope of } SQ = m_2 &= \frac{b(e - \cos \theta)}{e \sin \theta} \bigg/ \left(\frac{a}{e} - ae \right) \\ &= \frac{b(e - \cos \theta)}{a(1 - e^2) \sin \theta} \end{aligned}$$

Thus, $m_1 \times m_2$

$$\begin{aligned} &= \frac{b \sin \theta}{a(\cos \theta - e)} \times \frac{b(e - \cos \theta)}{a(1 - e^2) \sin \theta} \\ &= -\frac{b^2}{a^2(1 - e^2)} = -\frac{b^2}{b^2} = -1 \end{aligned}$$

Hence, the result.

32. Any tangent to the ellipse is

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1 \quad \dots(i)$$

Its point of contact is $P(a \cos \theta, b \sin \theta)$ and its slope is

$$-\frac{b}{a} \cot \theta$$

Also the focus is $S(ae, 0)$.

Any line through the focus S and perpendicular to the tangent (i) is

$$y - 0 = \frac{a}{b} \tan \theta (x - ae) \quad \dots(ii)$$

Also the equation of CP is

$$y - 0 = \frac{a}{b} \tan \theta (x - 0) \quad \dots(iii)$$

Eliminating θ between Eqs (ii) and (iii), we get

$$\left(\frac{a^2}{b^2}\right) \left(\frac{x - ae}{x}\right) = 1$$

$$\Rightarrow \left(\frac{x - ae}{x}\right) = \left(\frac{b^2}{a^2}\right)$$

$$\Rightarrow \left(1 - \frac{ae}{x}\right) = \left(\frac{b^2}{a^2}\right)$$

$$\Rightarrow \left(1 - \frac{b^2}{a^2}\right) = \left(\frac{ae}{x}\right)$$

$$\Rightarrow \left(1 - \frac{a^2(1 - e^2)}{a^2}\right) = \left(\frac{ae}{x}\right)$$

$$\Rightarrow e^2 = \left(\frac{ae}{x}\right)$$

$$\Rightarrow x = \frac{a}{e}$$

Hence, the result.

33. Consider a point P on the ellipse whose coordinates are

$$(a \cos \theta, b \sin \theta)$$

The equation of the tangent at P is

$$\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$$

Since the tangent makes equal angles with the axes, so its slope is ± 1 .

$$\text{Thus, } -\frac{b \cos \theta}{a \sin \theta} = \pm 1$$

$$\Rightarrow \frac{b^2 \cos^2 \theta}{a^2 \sin^2 \theta} = 1$$

$$\Rightarrow \frac{\sin^2 \theta}{a^2} = \frac{\cos^2 \theta}{b^2} = \frac{1}{a^2 + b^2}$$

$$\Rightarrow \sin^2 \theta = \frac{a^2}{a^2 + b^2} \text{ and } \cos^2 \theta = \frac{b^2}{a^2 + b^2}$$

Hence the point P is

$$\left(\pm \frac{a^2}{\sqrt{a^2 + b^2}}, \pm \frac{b^2}{\sqrt{a^2 + b^2}}\right)$$

34. As we know that the locus of the point of intersection of two perpendicular tangents is the director circle.

Hence, the equation of the director circle to the given ellipse is

$$x^2 + y^2 = a^2 + b^2 = 16 + 9 = 25$$

$$\Rightarrow x^2 + y^2 = 25$$

35. As we know that the locus of the point of intersection of two perpendicular tangents to an ellipse is the director circle.

Hence, the equation of the director circle is

$$x^2 + y^2 = a^2 + b^2 = 4 + 1 = 5$$

$$\Rightarrow x^2 + y^2 = 5$$

which is the required locus of P .

36. The equations of the pair of tangents to the ellipse

$$2x^2 + 3y^2 = 1 \text{ from the point } (1, 1) \text{ is}$$

$$(2x^2 + 3y^2 - 1)(2 + 3 - 1) = (2x + 3y - 1)^2$$

$$\Rightarrow 4(2x^2 + 3y^2 - 1) = (2x + 3y - 1)^2$$

$$\Rightarrow 4(2x^2 + 3y^2 - 1) = 4x^2 + 9y^2 + 1 + 12xy - 4x - 6y$$

$$\Rightarrow 4x^2 - 3y^2 - 12xy - 4x + 6y - 3 = 0$$

37. The equation of the given ellipse is

$$3x^2 + 2y^2 = 5$$

$$\Rightarrow \frac{x^2}{5/3} + \frac{y^2}{5/2} = 1$$

The equation of any tangents to the ellipse are

$$y = mx \pm \sqrt{b^2 m^2 + a^2}$$

which is passing through $(1, 2)$, so

$$2 = m \pm \sqrt{\frac{5}{2} m^2 + \frac{5}{3}}$$

$$\Rightarrow (2 - m)^2 = \frac{5}{2} m^2 + \frac{5}{3}$$

$$\Rightarrow (2 - m)^2 = \frac{15m^2 + 6}{5}$$

$$\Rightarrow 9m^2 + 24m - 14 = 0$$

Let its roots are m_1 and m_2 .

$$\text{Then } m_1 + m_2 = -24/9 \text{ and } m_1 m_2 = -14/9$$

Let θ be the angle between them.

Then

$$\begin{aligned} \tan(\theta) &= \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right| \\ &= \frac{\sqrt{(m_1 + m_2)^2 - 4m_1 m_2}}{1 + m_1 m_2} \end{aligned}$$

$$= \frac{\sqrt{\frac{64}{9} + \frac{56}{9}}}{1 - \frac{14}{9}} = \frac{\sqrt{120}}{3} \times \frac{9}{5} = 12 \times \sqrt{\frac{5}{6}}$$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{12\sqrt{6}}{\sqrt{5}} \right)$$

38. Do yourself.

39. Do yourself.

40. Do yourself.

41. Do yourself.

42. Let one end of a latus rectum is $L \left(ae, \frac{b^2}{a} \right)$ and minor axis be $B(0, -b)$.

The equation of the normal to the ellipse at L is

$$\frac{a}{e}x - ay = a^2 - b^2$$

which is passing through $B'(0, -b)$, so we have

$$ab = a^2 - b^2$$

$$\Rightarrow ab = a^2 - a^2(1 - e^2) = a^2 e^2$$

$$\Rightarrow b = ae^2$$

$$\Rightarrow b^2 = a^2 e^4$$

$$\Rightarrow a^2(1 - e^2) = a^2 e^4$$

$$\Rightarrow e^4 + e^2 - 1 = 0$$

Hence, the result.

43. Let the ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $P(a \cos \theta, b \sin \theta)$ be any point on the ellipse.

Then,

$$SP = a - ex = a - ae \cos \theta = a(1 - e \cos \theta)$$

and

$$SP = a + ex = a + ae \cos \theta = a(1 + e \cos \theta)$$

The equation of the normal at P is

$$ax \sec \theta - by \operatorname{cosec} \theta = a^2 - b^2$$

Now,

$$L = \left(\frac{a^2 - b^2}{a} \cos \theta, 0 \right) = (ae^2 \cos \theta, 0)$$

$$\Rightarrow OL = ae^2 \cos \theta$$

$$\text{Thus, } S'L = ae + ae^2 \cos \theta = a^3(1 + e \cos \theta),$$

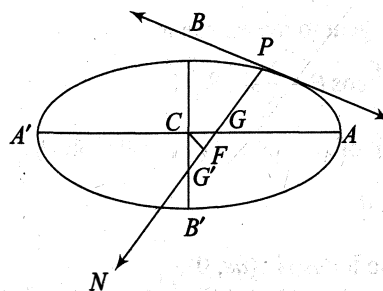
$$SL = ae - ae^2 \cos \theta = ae(1 - e \cos \theta)$$

$$\text{Now, } \frac{S'P}{SP} = \frac{1 + e \cos \theta}{1 - e \cos \theta} = \frac{S'L}{SL}$$

$\Rightarrow PL$ bisects the $\angle S'PS$ internally.

Since $PL \perp PT$, therefore, PT will bisect $\angle S'PS$ externally.

44. Let the ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $P(a \cos \theta, b \sin \theta)$ be any point on the ellipse.



The equation of the normal at P is

$$ax \sec \theta - by \operatorname{cosec} \theta = a^2 - b^2 \quad \dots(i)$$

Since the line (i) meets the major axis at G and minor axis at G' respectively, then

$$G = \left(\frac{a^2 - b^2}{a} \cos \theta, 0 \right)$$

$$\text{and } E = \left(0, -\frac{a^2 - b^2}{a} \sin \theta \right)$$

$PF = CQ =$ length of perpendicular from $C(0, 0)$ on the tangent $\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$ at P .

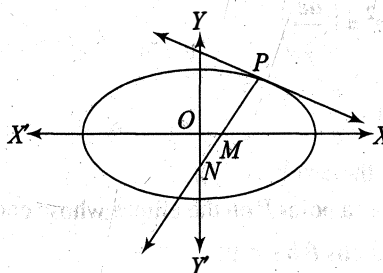
$$= \frac{ab}{\sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}}$$

$$\text{Also, } PG = \frac{b}{a} \sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}$$

$$\text{and } PG' = \frac{a}{b} \sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}$$

$$\text{Thus, } PF \cdot PG = b^2 \text{ and } PF \cdot PG' = a^2.$$

45. The equation of the normal to the ellipse at $P(\theta)$ is $ax \sec \theta - by \operatorname{cosec} \theta = a^2 - b^2$



Thus the co-ordinates of M and N are

$$\left(0, \left(\frac{b^2 - a^2}{b} \right) \sin \theta \right)$$

$$\text{and } \left(\left(\frac{a^2 - b^2}{a} \right) \cos \theta, 0 \right)$$

$$PM = \sqrt{a^2 \cos^2 \theta + \left(\frac{b^2 - a^2}{b} - b \right)^2 \sin^2 \theta}$$

$$= \sqrt{a^2 \cos^2 \theta + \frac{a^4}{b^2} \sin^2 \theta}$$

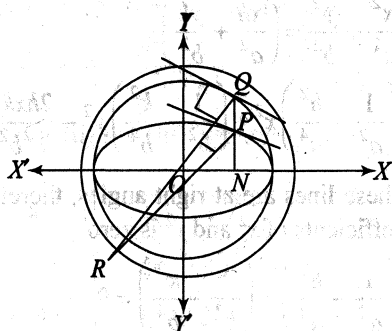
$$= a^2 \sqrt{\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2}}$$

$$\begin{aligned}\text{Also, } PN &= \sqrt{\left(a - \frac{a^2 - b^2}{a}\right)^2 \cos^2 \theta + b^2 \sin^2 \theta} \\ &= \sqrt{\frac{b^4 \cos^2 \theta}{a^2} + b^2 \sin^2 \theta} \\ &= b^2 \sqrt{\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2}}\end{aligned}$$

Thus, $PM : PN = a^2 : b^2$

Hence, the result.

46. Since the ordinate P meets the circle at Q , the co-ordinates of P and Q are $(a \cos \theta, b \sin \theta)$ and $(a \cos \theta, a \sin \theta)$, respectively.



The equation of the normal to the ellipse at $P(\theta)$ is

$$ax \sec \theta - by \operatorname{cosec} \theta = a^2 - b^2 \quad \dots(i)$$

and the equation of normal to the auxiliary circle at $Q(a \cos \theta, a \sin \theta)$ is

$$y = (\tan \theta)x \quad \dots(ii)$$

Solving, we get

$$\cos \theta = \frac{x}{(a+b)} \text{ and } \sin \theta = \frac{y}{(a+b)}$$

Squaring and adding, we get

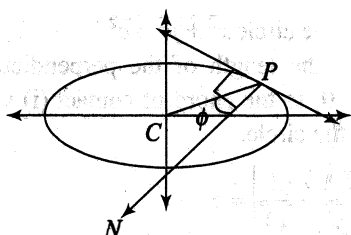
$$x^2 + y^2 = (a+b)^2$$

47. The equation of any normal at $P(\theta)$ is

$$x \sec \theta - by \operatorname{cosec} \theta = a^2 - b^2.$$

$$\therefore \text{Slope of normal} = m_1 = \frac{a}{b} \tan \theta$$

$$\text{and slope of } CP = m_2 = \frac{b}{a} \tan \theta.$$



Let ϕ be the angle between CP and the normal at P .

$$\begin{aligned}\therefore \tan \phi &= \left(\frac{\frac{a}{b} \tan \theta - \frac{b}{a} \tan \theta}{1 + \tan^2 \theta} \right) \\ &= \left(\frac{a^2 - b^2}{ab} \right) \frac{\tan \theta}{\sec^2 \theta} \\ &= \left(\frac{a^2 - b^2}{2ab} \right) \times \sin 2\theta\end{aligned}$$

Thus, it is maximum when

$$2\theta = 90^\circ \Rightarrow \theta = 45^\circ.$$

Therefore, the maximum value is

$$\left(\frac{a^2 - b^2}{2ab} \right)$$

48. The equation of any normal to the ellipse at $P(\theta)$ is

$$ax \sec \theta - by \operatorname{cosec} \theta = a^2 - b^2 \quad \dots(i)$$

Let C be the centre of the ellipse.

Then CM = Length of perpendicular from the centre C to the normal

$$\begin{aligned}&= \frac{(a^2 - b^2)}{\sqrt{a^2 \sec^2 \theta + b^2 \operatorname{cosec}^2 \theta}} \\ &= \frac{(a^2 - b^2)}{\sqrt{a^2 + b^2 + a^2 \tan^2 \theta + b^2 \cot^2 \theta}} \\ &< \frac{(a^2 - b^2)}{\sqrt{a^2 + b^2 + 2ab}} = a - b\end{aligned}$$

49. The equation of tangent at P is

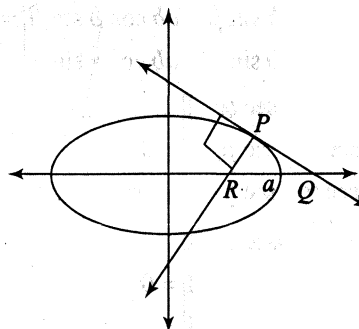
$$\frac{x}{a} \cos \phi + \frac{y}{b} \sin \phi = 1 \quad \dots(i)$$

and the equation of normal at P is

$$ax \sec \phi - by \operatorname{cosec} \phi = a^2 - b^2 \quad \dots(ii)$$

$$\therefore Q = (a \sec \phi, 0)$$

$$\text{and } R = \left(\frac{(a^2 - b^2) \cos \phi}{a}, 0 \right)$$



Therefore,

$$QR = a$$

$$\Rightarrow a \sec \phi - \frac{(a^2 - b^2) \cos \phi}{a} = a$$

$$\Rightarrow a^2 \sin^2 \phi + b^2 \cos^2 \phi = a^2 \cos \phi$$

$$\begin{aligned} \Rightarrow a^2 \sin^2 \phi + a^2(1 - e^2) \cos^2 \phi &= a^2 \cos \phi \\ \Rightarrow a^2(\sin^2 + \cos^2) - a^2 e^2 \cos^2 \phi &= a^2 \cos \phi \\ \Rightarrow a^2 - a^2 e^2 \cos^2 \phi &= a^2 \cos \phi \\ \Rightarrow c^2 \cos^2 \phi + \cos \phi - 1 &= 0 \end{aligned}$$

50. The equation of the normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at $P(x_1, y_1)$, $Q(x_2, y_2)$ and $R(x_3, y_3)$ are

$$\frac{a^2 x}{x_1} - \frac{b^2 y}{y_1} = a^2 - b^2 \quad \dots(i)$$

$$\frac{a^2 x}{x_2} - \frac{b^2 y}{y_2} = a^2 - b^2 \quad \dots(ii)$$

$$\text{and } \frac{a^2 x}{x_3} - \frac{b^2 y}{y_3} = a^2 - b^2 \quad \dots(iii)$$

Eliminating x and y from Eqs (i), (ii) and (iii), we get

$$\begin{vmatrix} \frac{a^2}{x_1} & -\frac{b^2}{y_1} & (a^2 - b^2) \\ \frac{a^2}{x_2} & -\frac{b^2}{y_2} & (a^2 - b^2) \\ \frac{a^2}{x_3} & -\frac{b^2}{y_3} & (a^2 - b^2) \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} \frac{1}{x_1} & \frac{1}{y_1} & 1 \\ \frac{1}{x_2} & \frac{1}{y_2} & 1 \\ \frac{1}{x_3} & \frac{1}{y_3} & 1 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x_1 & y_1 & x_1 y_1 \\ x_2 & y_2 & x_2 y_2 \\ x_3 & y_3 & x_3 y_3 \end{vmatrix} = 0$$

Also

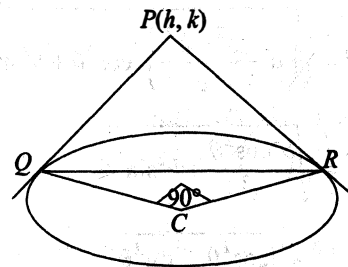
$$\begin{vmatrix} a \cos \alpha & b \sin \alpha & ab \cos \alpha \sin \alpha \\ a \cos \beta & b \sin \beta & ab \cos \beta \sin \beta \\ a \cos \gamma & b \sin \gamma & ab \cos \gamma \sin \gamma \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} \operatorname{cosec} \alpha & \sec \alpha & 1 \\ \operatorname{cosec} \beta & \sec \beta & 1 \\ \operatorname{cosec} \gamma & \sec \gamma & 1 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} \sec \alpha & \operatorname{cosec} \alpha & 1 \\ \sec \beta & \operatorname{cosec} \beta & 1 \\ \sec \gamma & \operatorname{cosec} \gamma & 1 \end{vmatrix} = 0$$

Hence the result.

51. Let the point from which tangents are drawn be (h, k) . The equation of the chord of contact from the point (h, k) to the given ellipse is $\frac{hx}{a^2} + \frac{ky}{b^2} = 1$.



It subtends a right angle at the centre $(0, 0)$ of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \dots(ii)$$

From Eqs (i) and (ii), we get

$$\begin{aligned} \frac{x^2}{a^2} + \frac{y^2}{b^2} &= \left(\frac{xh}{a^2} + \frac{yk}{b^2} \right)^2 \\ \Rightarrow \left(\frac{1}{a^2} - \frac{h^2}{a^4} \right) x^2 + \left(\frac{1}{b^2} - \frac{k^2}{b^4} \right) y^2 - \frac{2hxy}{a^2 b^2} &= 0 \end{aligned}$$

Since these lines are at right angles, therefore sum of the co-efficients of x^2 and y^2 is zero.

$$\Rightarrow \left(\frac{1}{a^2} - \frac{h^2}{a^4} \right) + \left(\frac{1}{b^2} - \frac{k^2}{b^4} \right) = 0$$

$$\Rightarrow \left(\frac{h^2}{a^4} + \frac{k^2}{b^4} \right) = \frac{1}{a^2} + \frac{1}{b^2}$$

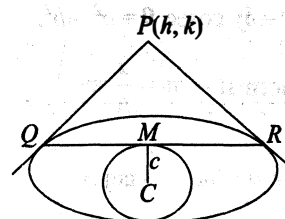
Hence, the locus of (h, k) is

$$\left(\frac{x^2}{a^4} + \frac{y^2}{b^4} \right) = \frac{1}{a^2} + \frac{1}{b^2}$$

52. Let the point from which the tangents are drawn be (h, k) .

So the equation of the chord of contact from the point (h, k) to the given ellipse is

$$\frac{hx}{a^2} + \frac{ky}{b^2} = 1 \quad \dots(i)$$



It touches the circle $x^2 + y^2 = c^2$.

Therefore, the length of the perpendicular from the centre $(0, 0)$ to the chord of contact (i) is equal to the radius of the circle.

$$\text{Thus, } \left| \frac{0+0-1}{\sqrt{\frac{h^2}{a^4} + \frac{k^2}{b^4}}} \right| = c$$

$$\Rightarrow \frac{h^2}{a^4} + \frac{k^2}{b^4} = \frac{1}{c^2}$$

Hence, the locus of (h, k) is

$$\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{c^2}$$

53. Let (x_1, y_1) be the point of intersection of the perpendicular tangents, so that (x_1, y_1) lies on the director circle

$$x_1^2 + y_1^2 = a^2 + b^2 \quad \dots(i)$$

The equation of the chord of contact from (x_1, y_1) is

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1 \quad \dots(ii)$$

Let its mid-point be (h, k) .

$\therefore$ The equation of the chord bisected at (h, k) is

$$\frac{hx}{a^2} + \frac{ky}{b^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} \quad \dots(iii)$$

From Eqs (ii) and (iii), we get

$$\frac{x_1}{h} = \frac{y_1}{k} = \frac{1}{\left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)}$$

Now from Eqs. (i), we get

$$\left(\frac{h}{\left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)}\right)^2 + \left(\frac{k}{\left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)}\right)^2 = (a^2 + b^2)$$

$$\Rightarrow (a^2 + b^2) \left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)^2 = (h^2 + k^2)$$

Hence, the locus of (h, k) is

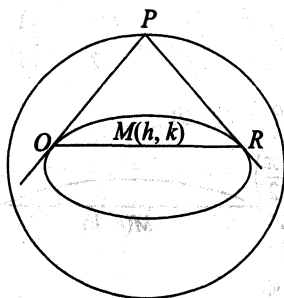
$$(a^2 + b^2) \left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^2 = (x^2 + y^2)$$

54. Let the point be $(c \cos \theta, c \sin \theta)$.

The equation of the tangent to the ellipse at $(c \cos \theta, c \sin \theta)$ is

$$\frac{x \cdot c \cos \theta}{a^2} + \frac{y \cdot c \sin \theta}{b^2} = 1 \quad \dots(i)$$

Let the co-ordinates of the mid-point be (h, k) .



$\therefore$ The equation of the chord bisected at (h, k) is

$$\frac{hx}{a^2} + \frac{ky}{b^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} \quad \dots(ii)$$

From Eqs (i) and (ii), we get

$$\frac{h}{c \cos \theta} = \frac{k}{c \sin \theta} = \frac{1}{\frac{h^2}{a^2} + \frac{k^2}{b^2}}$$

$$\Rightarrow \cos \theta = \frac{h}{c \left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)}$$

$$\text{and } \sin \theta = \frac{k}{c \left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)}$$

Squaring and adding, we get

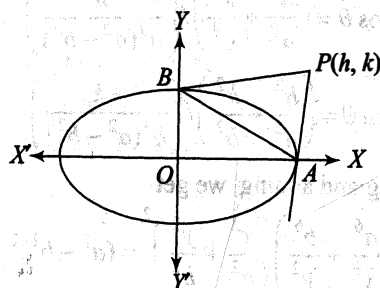
$$\left(\frac{h}{c \left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)}\right)^2 + \left(\frac{k}{c \left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)}\right)^2 = 1$$

$$\Rightarrow \left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)^2 = \left(\frac{c^2}{h^2 + k^2}\right)$$

Hence, the locus of (h, k) is

$$\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^2 = \left(\frac{c^2}{x^2 + y^2}\right)$$

55. Let the co-ordinates of P be (h, k) .



The equation of its chord of contact with respect to the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ is}$$

$$\frac{hx}{a^2} + \frac{ky}{b^2} = 1$$

It meets the axes in

$$A\left(\frac{a^2}{h}, 0\right) \text{ and } B\left(0, \frac{b^2}{k}\right)$$

$$\text{Now, area of the triangle } OAB = \frac{1}{2} \cdot OA \cdot OB$$

$$= \frac{1}{2} \cdot \frac{a^2}{h} \cdot \frac{b^2}{k}$$

$$= \frac{a^2 b^2}{2 \cdot h \cdot k}$$

$$= \text{constant} = \frac{\lambda}{2} \text{ (say)}$$

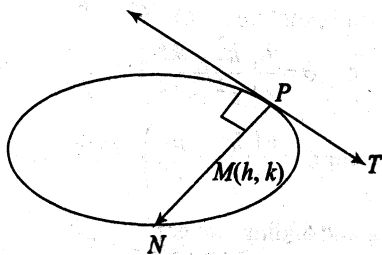
$$\text{Hence, the locus of } (h, k) \text{ is } x \cdot y = \frac{a^2 b^2}{\lambda}$$

which represents a hyperbola.

56. Let the point P be $(a \cos \theta, b \sin \theta)$.

$\therefore$ The equation of normal at P to the ellipse is

$$a \cdot x \cdot \sec \theta - b \cdot y \cdot \operatorname{cosec} \theta = a^2 - b^2 \quad \dots(i)$$



Let its mid-point be (h, k) .

$\therefore$ The equation of the chord bisected at (h, k) is

$$\frac{hx}{a^2} + \frac{ky}{b^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} \quad \dots(ii)$$

From Eqs (i) and (ii), we get

$$\frac{a \sec \theta}{(h/a^2)} = \frac{b \operatorname{cosec} \theta}{(-k/b^2)} = \frac{a^2 - b^2}{\left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)}$$

$$\Rightarrow \cos \theta = \left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right) / \left(\frac{h}{a^3(a^2 - b^2)}\right)$$

$$\text{and } \sin \theta = \left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right) / \left(\frac{-k}{b^3(a^2 - b^2)}\right)$$

Squaring and adding, we get

$$\left(\frac{a^6}{x^2} + \frac{b^6}{y^2}\right) \left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^2 = (a^2 - b^2)^2$$

57. Let the point be (h, k) .

The equation of the chord bisected at (h, k) to the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ is}$$

$$\frac{hx}{a^2} + \frac{ky}{b^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} \quad \dots(i)$$

and the equation of the tangent to the auxiliary circle $x^2 + y^2 = a^2$ at $(a \cos \theta, a \sin \theta)$ is

$$x \cos \theta + y \sin \theta = a \quad \dots(ii)$$

From Eqs (i) and (ii), we get

$$\frac{\cos \theta}{h/a^2} = \frac{\sin \theta}{k/b^2} = \frac{a}{\left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)}$$

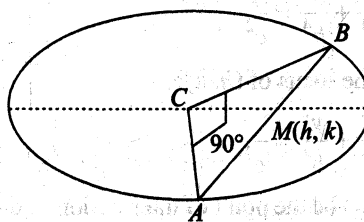
Squaring and adding, we get

$$a^2 \left(\frac{h^2}{a^4} + \frac{k^2}{b^4}\right) = \left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)^2$$

Hence the locus of (h, k) is

$$a^2 \left(\frac{x^2}{a^4} + \frac{y^2}{b^4}\right) = \left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^2$$

58. Let the point be (h, k) .



The equation of the chord bisected at (h, k) to the ellipse is

$$\frac{hx}{a^2} + \frac{ky}{b^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2}$$

$$\Rightarrow \left(\frac{\frac{hx}{a^2} + \frac{ky}{b^2}}{\left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)}\right) = 1$$

Now we make it a homogenous equation of 2nd degree.

$$\text{Thus, } \left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right) = \left(\frac{\frac{hx}{a^2} + \frac{ky}{b^2}}{\left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)}\right)^2$$

$$\Rightarrow \left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right) \left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)^2 = \left(\frac{hx}{a^2} + \frac{ky}{b^2}\right)^2$$

$$= \frac{h^2 x^2}{a^4} + \frac{k^2 y^2}{b^4} + \frac{2hky}{a^2 b^2}$$

Since the chord subtends right angle at the centre, so co-efficient of $x^2 +$ co-efficient of $y^2 = 0$

$$\frac{\alpha^2 x^2}{a^4} + \frac{\beta^2 y^2}{b^4} = \left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^2$$

$$\Rightarrow \left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)^2 \left(\frac{1}{a^2} + \frac{1}{b^2}\right) = \left(\frac{h^2}{a^4} + \frac{k^2}{b^4}\right)$$

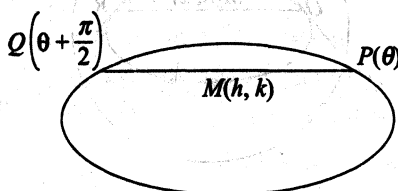
Hence, the locus of (h, k) is

$$\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^2 \left(\frac{1}{a^2} + \frac{1}{b^2}\right) = \left(\frac{x^2}{a^4} + \frac{y^2}{b^4}\right)$$

59. Let the point P be $(a \cos \theta, b \sin \theta)$ and Q be

$$\left(a \cos \left(\frac{\pi}{2} + \theta\right), b \sin \left(\frac{\pi}{2} + \theta\right)\right),$$

i.e. $(-a \sin \theta, b \cos \theta)$.



Let $M(h, k)$ be the mid-point of PQ .

Then, $h = \frac{a}{2}(\cos \theta - \sin \theta)$

and $k = \frac{b}{2}(\cos \theta + \sin \theta)$

$\Rightarrow \frac{2h}{a} = (\cos \theta - \sin \theta)$ and $\frac{2k}{b} = (\cos \theta + \sin \theta)$

Squaring and adding, we get

$$\left(\frac{4h^2}{a^2} + \frac{4k^2}{b^2}\right) = 2$$

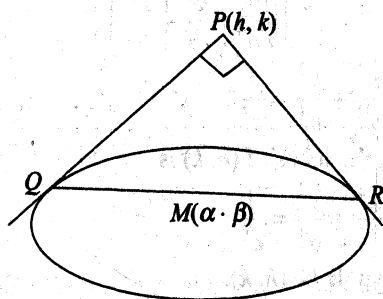
$\Rightarrow \left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right) = \frac{1}{2}$

Hence, the locus of (h, k) is

$$\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right) = \frac{1}{2}$$

60. Equation of the chord of contact to the tangents at (h, k) is

$$\frac{hx}{a^2} + \frac{ky}{b^2} = 1 \quad \dots(i)$$



The equation of the chord of the ellipse whose mid-point (α, β) is

$$\frac{\alpha x}{a^2} + \frac{\beta y}{b^2} = \frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2} \quad \dots(ii)$$

Since Eqs (i) and (ii) are the same, therefore

$$\frac{h}{\alpha} = \frac{k}{\beta} = \frac{1}{\left(\frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2}\right)}$$

$\Rightarrow h = \frac{\alpha}{\left(\frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2}\right)}$

and $k = \frac{\beta}{\left(\frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2}\right)}$

Also, (h, k) lies on the director circle of the given ellipse $x^2 + y^2 = a^2 + b^2$.

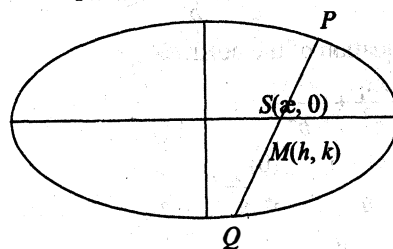
Thus, $h^2 + k^2 = a^2 + b^2$

$\Rightarrow \left(\frac{\alpha^2 + \beta^2}{a^2 + b^2}\right) = \left(\frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2}\right)^2$

Hence, the locus of (α, β) is

$$\left(\frac{x^2 + y^2}{a^2 + b^2}\right) = \left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^2$$

61. Let the mid-point be (h, k) .



The equation of the chord of the ellipse whose mid-point (h, k) is

$$\frac{hx}{a^2} + \frac{ky}{b^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2}$$

which passes through the focus $(ae, 0)$.

Thus, $\frac{hae}{a^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2}$

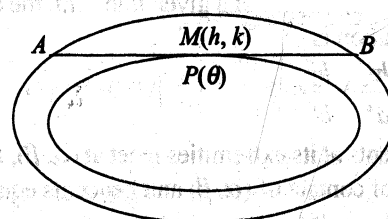
$\Rightarrow \frac{he}{a} = \frac{b^2 h^2 + a^2 k^2}{a^2 b^2}$

$\Rightarrow b^2 h^2 + a^2 k^2 = ab^2 he$

Hence, the locus of (h, k) is

$$b^2 h^2 + a^2 k^2 = ab^2 xe$$

62. Let the point be (h, k) .



The equation of the chord of the ellipse whose middle point (h, k) is

$$\frac{hx}{a^2} + \frac{ky}{b^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} \quad \dots(i)$$

and the equation of the tangent to the ellipse

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at $(\alpha \cos \theta, \beta \sin \theta)$ is

$$\frac{x}{\alpha} \cos \theta + \frac{y}{\beta} \sin \theta = 1$$

Since both the equations are identical, so

$$\frac{\cos \theta / \alpha}{h/a^2} = \frac{\sin \theta / \beta}{k/b^2} = \frac{1}{\left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)}$$

Squaring and adding, we get

$$\frac{\alpha^2 h^2}{a^4} + \frac{\beta^2 k^2}{b^4} = \left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)^2$$

Hence the locus of (h, k) is

$$\frac{\alpha^2 x^2}{a^4} + \frac{\beta^2 y^2}{b^4} = \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right)^2$$

63. Let the ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and its focus is $(ae, 0)$.

The equation of the polar is

$$\begin{aligned} \frac{xx_1}{a^2} + \frac{yy_1}{b^2} &= 1 \\ \Rightarrow \frac{x(ae)}{a^2} + \frac{y(0)}{b^2} &= 1 \\ \Rightarrow x &= \frac{a}{e} \end{aligned}$$

which is the directrix.

64. Let the pole be (x_1, y_1) .

The equation of the polar is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$ which is identical with $lx + my + n = 0$

$$\begin{aligned} \text{So, } \frac{x_1/a^2}{l} &= \frac{y_1/b^2}{m} = \frac{-1}{n} \\ \Rightarrow x_1 &= -\frac{a^2 l}{n}, y_1 = -\frac{b^2 m}{n} \end{aligned}$$

Hence, the pole is $\left(-\frac{a^2 l}{n}, -\frac{b^2 m}{n} \right)$

65. If (h, k) be the pole of a given line w.r.t. the ellipse, then its equation is

$$\frac{hx}{a^2} + \frac{ky}{b^2} = 1 \quad \dots(i)$$

If tangents at its extremities meet at (α, β) , then it is the chord of contact of (α, β) and hence its equation is

$$\frac{\alpha x}{a^2} + \frac{\beta y}{b^2} = 1 \quad \dots(ii)$$

Comparing Eqs (i) and (ii), we get

$$\begin{aligned} \frac{h/a^2}{\alpha/a^2} &= \frac{k/b^2}{\beta/b^2} = 1 \\ \Rightarrow h &= \alpha, k = \beta \end{aligned}$$

Thus, the pole (h, k) is the same as (α, β) , i.e. the intersection of tangents.

66. Let the pole be (h, k) .

The equation of polar w.r.t. the ellipse $x^2 + 4y^2 = 4$ is

$$hx + 4ky = 4$$

which is identical with $x + 4y = 4$

$$\begin{aligned} \text{So, } \frac{h}{1} &= \frac{4k}{4} = \frac{4}{4} \\ \Rightarrow \frac{h}{1} &= \frac{k}{1} = 1 \end{aligned}$$

Hence, the pole is $(1, 1)$

67. Let the pole be (h, k) .

The equation of polar w.r.t. the ellipse $c^2 x^2 + d^2 y^2 = 1$ is

$$\begin{aligned} c^2 hx + d^2 ky &= 1 \\ \Rightarrow d^2 ky &= -c^2 hx + 1 \end{aligned}$$

$$\Rightarrow y = -\frac{c^2 h}{d^2 k} x + \frac{1}{d^2 k}$$

which is a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\text{So, } c^2 = a^2 m^2 + b^2$$

$$\Rightarrow \frac{1}{(d^2 k)^2} = a^2 \left(-\frac{c^2 h}{d^2 k} \right)^2 + b^2$$

$$\Rightarrow a^2 c^4 h^2 + b^2 d^4 k^2 = 1$$

Hence, the locus of (h, k) is

$$a^2 c^4 x^2 - b^2 d^4 y^2 = 1$$

68. Let the pole be (h, k) .

The equation of the polar is $\frac{hx}{a^2} + \frac{ky}{b^2} = 1$

$$\text{It is given that } \left| \frac{0+0-1}{\sqrt{\frac{h^2}{a^4} + \frac{k^2}{b^4}}} \right| = c$$

$$\Rightarrow \frac{h^2}{a^4} + \frac{k^2}{b^4} = \frac{1}{c^2}$$

Hence, the locus of (h, k) is

$$\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{c^2}$$

69. Let the pole be (h, k) .

The equation of the polar is $\frac{hx}{a^2} + \frac{ky}{b^2} = 1$... (i)

and the equation of the normal to the ellipse is

$$ax \sec \phi - by \csc \phi = a^2 - b^2 \quad \dots(ii)$$

Comparing Eqs (i) and (ii), we get

$$\begin{aligned} \frac{h/a^2}{a \sec \phi} &= -\frac{k/b^2}{b \csc \phi} = \frac{1}{a^2 - b^2} \\ \Rightarrow \frac{h \cos \phi}{a^3} &= -\frac{k \sin \phi}{b^3 \csc \phi} = \frac{1}{a^2 - b^2} \end{aligned}$$

$$\Rightarrow \cos \phi = \frac{a^3}{h(a^2 - b^2)}, \sin \phi = -\frac{b^3}{k(a^2 - b^2)}$$

Eliminating ϕ , we get

$$\begin{aligned} \left(\frac{a^3}{h(a^2 - b^2)} \right)^2 + \left(-\frac{b^3}{k(a^2 - b^2)} \right)^2 &= 1 \\ \Rightarrow \frac{a^6}{h^2} + \frac{b^6}{k^2} &= (a^2 - b^2)^2 \end{aligned}$$

Hence, the locus of (h, k) is $\frac{a^6}{x^2} + \frac{b^6}{y^2} = (a^2 - b^2)^2$

70. Let the pole be (h, k)

The equation of the polar w.r.t. the parabola is

$$yk = 2a(x + h)$$

$$\Rightarrow yk = 2ax + 2ah$$

$$\Rightarrow y = \frac{2a}{k}x + \frac{2ah}{k}$$

which is a tangent to the ellipse $\frac{x^2}{\alpha^2} + \frac{y^2}{\beta^2} = 1$

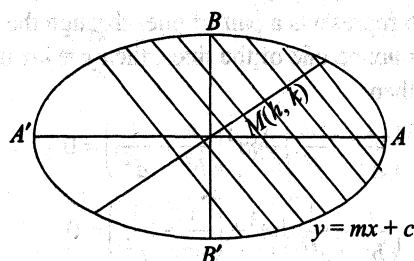
$$\text{So, } \left(\frac{2ah}{k}\right)^2 = \alpha^2 \left(\frac{2a}{k}\right)^2 + \beta^2$$

$$\Rightarrow 4a^2h^2 = 4a^2\alpha^2 + k^2\beta^2$$

Hence, the locus of pole (h, k) is

$$4a^2x^2 = y^2\beta^2 + 4a^2\alpha^2$$

71.



Let (h, k) be the mid-point of the chord $y = mx + c$ of the

ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Then $T = S_1$

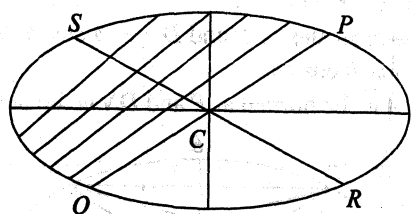
$$\Rightarrow \frac{xh}{a^2} + \frac{yk}{b^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2}$$

$$\Rightarrow k = -\frac{b^2h}{a^2m}$$

Hence, the locus of the mid-point is $y = -\frac{b^2x}{a^2m}$

72. Two diameters are said to be conjugate when each bisects all chords parallel to the other.

If $y = m_1x$ and $y = m_2x$ be two conjugate diameters of an ellipse, then $m_1m_2 = -\frac{b^2}{a^2}$.



Let PQ and RS be two conjugate diameters. Then the co-ordinates of the four extremities of two conjugate diameters are

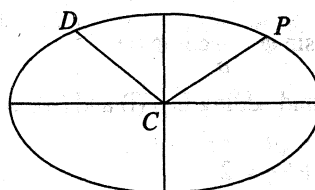
$$P(a \cos \phi, b \sin \phi),$$

$$Q(-a \cos \phi, b \sin \phi),$$

$$S(-a \sin \phi, b \cos \phi)$$

$$\text{and } R(a \sin \phi, -b \cos \phi)$$

73.



Let CP and CD be two conjugate semi-diameters of an

ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and let the eccentric angle of P is ϕ . Thus the eccentric angle of D is $\frac{\pi}{2} + \phi$.

Therefore the co-ordinates of P and D are $(a \cos \phi, b \sin \phi)$ and

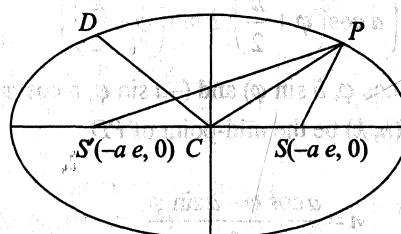
$$\left(a \cos \left(\frac{\pi}{2} + \phi\right), b \sin \left(\frac{\pi}{2} + \phi\right)\right)$$

i.e. $(-a \sin \phi, b \cos \phi)$

Thus $CP^2 + CD^2$

$$= (a^2 \cos^2 \phi + b^2 \sin^2 \phi) + (a^2 \sin^2 \phi + b^2 \cos^2 \phi) \\ = a^2 + b^2$$

74.



Let CP and CD be the conjugate diameters of the ellipse.

Let $P = (a \cos \phi, b \sin \phi)$. then the co-ordinates of D is $(-a \sin \phi, b \cos \phi)$.

Thus,

$$SP \cdot S'P = (a - ae \cos \phi)(a + ae \cos \phi) \\ = a^2 - a^2e^2 \cos^2 \phi \\ = a^2 - (a^2 - b^2) \cos^2 \phi \\ = a^2 \sin^2 \phi + b^2 \cos^2 \phi \\ = CD^2$$

75 Let the eccentric angle of P is (ϕ) and the eccentric angle of M is $\left(\phi + \frac{\pi}{2}\right)$.

Then the co-ordinates of P and M are $(a \cos \phi, b \sin \phi)$ and

$$\left(a \cos \left(\phi + \frac{\pi}{2}\right), b \sin \left(\phi + \frac{\pi}{2}\right)\right)$$

i.e. $(a \cos \phi, b \sin \phi)$ and $(-a \sin \phi, b \cos \phi)$

The equation of the tangent at P is

$$\frac{x}{a} \cos \phi + \frac{y}{b} \sin \phi = 1 \quad \dots(i)$$

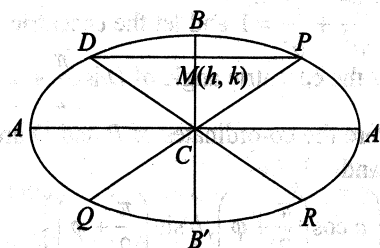
and the equation of the tangent at M is

$$\frac{x}{-a} \sin \phi + \frac{y}{b} \cos \phi = 1 \quad \dots(ii)$$

Squaring and adding Eqs (i) and (ii), we get

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$$

76. Let the equation of the ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.



Let the eccentric angle of P is (ϕ) and the eccentric angle of D is $(\phi + \frac{\pi}{2})$.

Then the co-ordinates of P and D are $(a \cos \phi, b \sin \phi)$ and

$$(-a \sin \phi, b \cos \phi)$$

i.e. $(a \cos \phi, b \sin \phi)$ and $(-a \sin \phi, b \cos \phi)$

Let $M(h, k)$ be the mid-point of PD .

Then

$$h = \frac{a \cos \phi - a \sin \phi}{2}$$

$$\text{and } k = \frac{a \cos \phi + b \sin \phi}{2} \Rightarrow \frac{(2h)^2}{a^2} + \frac{(2k)^2}{b^2} = (\cos \phi - \sin \phi)^2 + (\cos \phi + \sin \phi)^2$$

$$\Rightarrow \frac{h^2}{a^2} + \frac{k^2}{b^2} = 1$$

Hence, the locus of (h, k) is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

77. The equation of the given diameter is

$$ax - by = 0$$

$$\Rightarrow y = \frac{a}{b}x \quad \dots(i)$$

Thus, $m_1 = \frac{a}{b}$

Let the diameter conjugate to (i) be $y = m_2x$

As we know that, two diameters $y = m_1x$ and $y = m_2x$

are conjugate, if $m_1 \cdot m_2 = -\frac{b^2}{a^2}$

$$\Rightarrow \frac{a}{b} \times m_2 = -\frac{b^2}{a^2}$$

$$\Rightarrow m_2 = -\frac{b^3}{a^3}$$

Hence, the required diameters is $y = -\frac{b^3}{a^3}x$

$$\Rightarrow a^3y + b^3x = 0$$

78. The given ellipses are

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \dots(i)$$

$$\text{and } \frac{x^2}{c^2} + \frac{y^2}{d^2} = 1 \quad \dots(ii)$$

The equation of lines passing through the point of intersection of the ellipses are

$$x^2 \left(\frac{1}{a^2} - \frac{1}{c^2} \right) + y^2 \left(\frac{1}{b^2} - \frac{1}{d^2} \right) = 0 \quad \dots(iii)$$

which represents a pair of lines through the origin.

If $y = mx$ be one of the lines, then $y = mx$ must satisfy (iii), then

$$\left(\frac{1}{a^2} - \frac{1}{c^2} \right) + m^2 \left(\frac{1}{b^2} - \frac{1}{d^2} \right) = 0$$

$$\Rightarrow \left(\frac{1}{b^2} - \frac{1}{d^2} \right) m^2 + \left(\frac{1}{a^2} - \frac{1}{c^2} \right) = 0$$

which is a quadratic in m . Let it has roots m_1 and m_2 .

$$\text{Then, } m_1 m_2 = -\frac{\left(\frac{1}{a^2} - \frac{1}{c^2} \right)}{\left(\frac{1}{b^2} - \frac{1}{d^2} \right)}$$

$$\Rightarrow -\frac{b^2}{a^2} = \frac{\left(\frac{1}{a^2} - \frac{1}{c^2} \right)}{\left(\frac{1}{b^2} - \frac{1}{d^2} \right)}$$

$$\Rightarrow a^2 \left(\frac{1}{a^2} - \frac{1}{c^2} \right) = -b^2 \left(\frac{1}{b^2} - \frac{1}{d^2} \right)$$

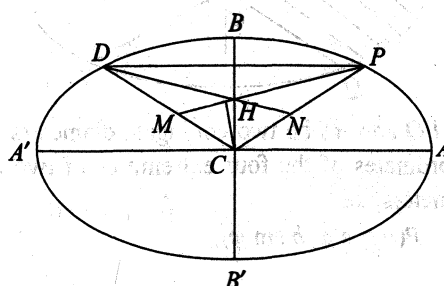
$$\Rightarrow \left(1 - \frac{a^2}{c^2} \right) = \left(-1 + \frac{b^2}{d^2} \right)$$

$$\Rightarrow \frac{a^2}{c^2} + \frac{b^2}{d^2} = 2$$

Hence, the result.

79. The co-ordinates of P and D are $(a \cos \phi, b \sin \phi)$ and $(-a \sin \phi, b \cos \phi)$

Let PM is the normal at P and DN is the normal at D .



The equations of the normals at P and D are

$$ax \sec \phi - by \operatorname{cosec} \phi = a^2 - b^2$$

and $-ax \operatorname{cosec} \phi - by \sec \phi = a^2 - b^2$

respectively.

Since, $H(\alpha, \beta)$ is the point of intersection of the normals, so

$$a\alpha \sec \phi - b\beta \operatorname{cosec} \phi - (a^2 - b^2) = 0 \quad \dots(i)$$

and

$$a\alpha \operatorname{cosec} \phi + b\beta \sec \phi + (a^2 - b^2) = 0 \quad \dots(ii)$$

Eliminating ϕ from Eqs (i) and (ii), we get

$$\frac{\sec \phi}{(a\alpha - b\beta)} = \frac{\operatorname{cosec} \phi}{-(a\alpha + b\beta)} = \frac{(a^2 - b^2)}{(a^2\alpha^2 + b^2\beta^2)}$$

$$\Rightarrow \cos \phi = \frac{(a^2\alpha^2 + b^2\beta^2)}{(a^2 - b^2)(a\alpha - b\beta)}$$

$$\text{and } \sin \phi = \frac{(a^2\alpha^2 + b^2\beta^2)}{-(a^2 - b^2)(a\alpha + b\beta)}$$

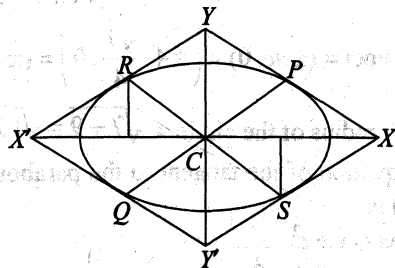
Squaring and adding, we get

$$2(a^2\alpha^2 + b^2\beta^2)^2 = (a^2 - b^2)^2(a^2\alpha^2 - b^2\beta^2)^2$$

Hence, the locus of $H(\alpha, \beta)$ is

$$2(a^2x^2 + b^2y^2)^3 = (a^2 - b^2)^2(a^2x^2 - b^2y^2)^2.$$

80.



Let PCQ and RCS be two conjugate diameters of the

ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Then the co-ordinates of P, Q, R , and S are

$P(a \cos \phi, b \sin \phi)$, $Q(-a \cos \phi, -b \sin \phi)$, $R(-a \sin \phi, b \cos \phi)$ and $S(a \sin \phi, -b \cos \phi)$ respectively.

The equations of tangents at P, R, Q and S are

$$\frac{x}{a} \cos \phi + \frac{y}{b} \sin \phi = 1,$$

$$-\frac{x}{a} \sin \phi + \frac{y}{b} \cos \phi = 1,$$

$$-\frac{x}{a} \cos \phi - \frac{y}{b} \sin \phi = 1,$$

$$\text{and } \frac{x}{a} \sin \phi - \frac{y}{b} \cos \phi = 1$$

Thus, the tangents at P and Q are parallel. Also the tangents at R and S are parallel. Hence, the tangents at P, R, Q, S form a parallelogram.

Area of the parallelogram $= MNM'N'$
 $= 4$ (the area of the parallelogram $CPMR$)

$$= 4 \times \sqrt{a^2 \cos^2 \phi + b^2 \sin^2 \phi} \times \frac{ab}{\sqrt{a^2 \cos^2 \phi + b^2 \sin^2 \phi}}$$

$$= 4ab$$

= constant

Hence, the result.

81. Let the eccentric angles at P and Q be ϕ and $\left(\frac{\pi}{2} + \phi\right)$ respectively

The equation of the tangents at P and Q

$$\text{are } \frac{x}{a} \cos \phi + \frac{y}{b} \sin \phi = 1$$

$$\text{and } -\frac{x}{a} \sin \phi + \frac{y}{b} \cos \phi = 1$$

respectively

Squaring and adding, we get

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$$

Hence, the result.

82. Thus, the equation of the ellipse is

$$y^2 + \frac{2}{3}x^2 = 1$$

$$\Rightarrow \frac{x^2}{3} + \frac{y^2}{2} = 1$$

Hence, the eccentricity is

$$e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{2}{3}} = \frac{1}{\sqrt{3}}$$

83. The given equation of an ellipse is

$$16x^2 + 25y^2 = 400$$

$$\Rightarrow \frac{x^2}{25} + \frac{y^2}{16} = 1$$

From the reflection property of an ellipse, the reflection ray passes through the focus.

Thus, the co-ordinates of the other focus = $(3, 0)$.

When $y = 4$, then $x = 0$.

So the point is $(0, 4)$.

So the equation of the reflection ray is

$$\frac{x}{3} + \frac{y}{4} = 1$$

$$\Rightarrow 4x + 3y = 12$$